# Coronavirus tweets NLP - Text Classification

**1. Dataset Name, Source, Version, and License**

**Dataset name:** Coronavirus tweets NLP - Text Classification dataset

**Source:** Coronavirus tweets NLP - Text Classification – Kaggle

**Author:** Aman Miglani (Owner)

**Version:** Not explicitly specified in the supplied files. The dataset version and access date will be recorded after verifying the original source.

**License:** Data files © Original Authors

The dataset contains tweets concerning COVID-19 and related public responses. Each tweet is assigned one of five sentiment labels, making the dataset suitable for supervised, multiclass text classification.

**2. Task Definition**

The objective is to classify a COVID-19-related tweet into one of five sentiment categories:

- Extremely Negative
- Negative
- Neutral
- Positive
- Extremely Positive

The task is formulated as a five-class text classification problem.

**Input:** The text in the OriginalTweet column.

**Target:** The corresponding label in the Sentiment column.

The remaining columns are UserName, ScreenName, Location, and TweetAt. These fields will be retained for dataset examination and descriptive analysis but excluded from the primary model input. In particular, identifiers and location information will not be used as features because they may introduce spurious correlations that do not reflect sentiment expressed in the tweet.

**3. Dataset Size and Preliminary Analysis**

The dataset consists of two supplied CSV files.

| Property | Training set | Test set |
|---|---|---|
| Number of samples | 41,157 | 3,798 |
| Number of columns | 6 | 6 |
| Number of target classes | 5 | 5 |
| Missing tweet texts | 0 | 0 |
| Missing sentiment labels | 0 | 0 |
| Missing locations | 8,590 | 834 |
| Date coverage | 16 Mar–14 Apr 2020 | 2–16 Mar 2020 |

The combined dataset contains **44,955 records**. All tweet texts and sentiment labels are present in both files, so no samples need to be removed due to missing input text or target labels. The missing Location values do not affect the proposed experiment because location will not be used as a model feature.

### 3.1. Class Distribution

The training set contains the following class distribution:

| Sentiment | Samples | Percentage |
|---|---|---|
| Positive | 11,422 | 27.75% |
| Negative | 9,917 | 24.10% |
| Neutral | 7,713 | 18.74% |
| Extremely Positive | 6,624 | 16.09% |
| Extremely Negative | 5,481 | 13.32% |
| **Total** | **41,157** | **100%** |

The test set contains 947 Positive, 1,041 Negative, 619 Neutral, 599 Extremely Positive, and 592 Extremely Negative samples.

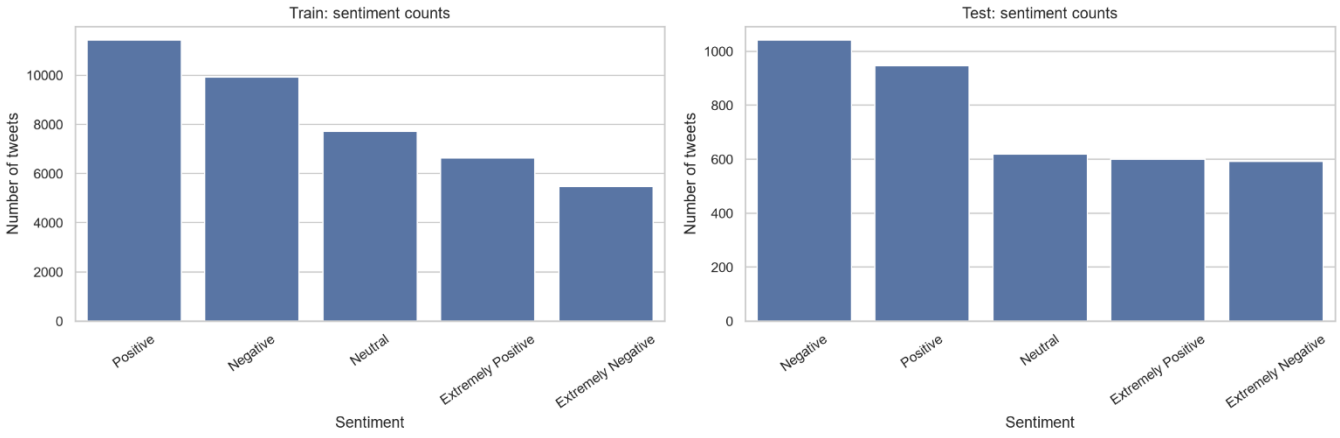
The dataset is moderately imbalanced: Positive is the most frequent training class, while Extremely Negative is the least frequent. However, no single class dominates the dataset.

Therefore, the experiment will report both accuracy and macro-averaged F1-score. Macro-F1 is particularly important because it gives each sentiment class equal weight regardless of its sample count.

### 3.2. Text Characteristics

The preliminary text analysis shows that tweets have the following characteristics:

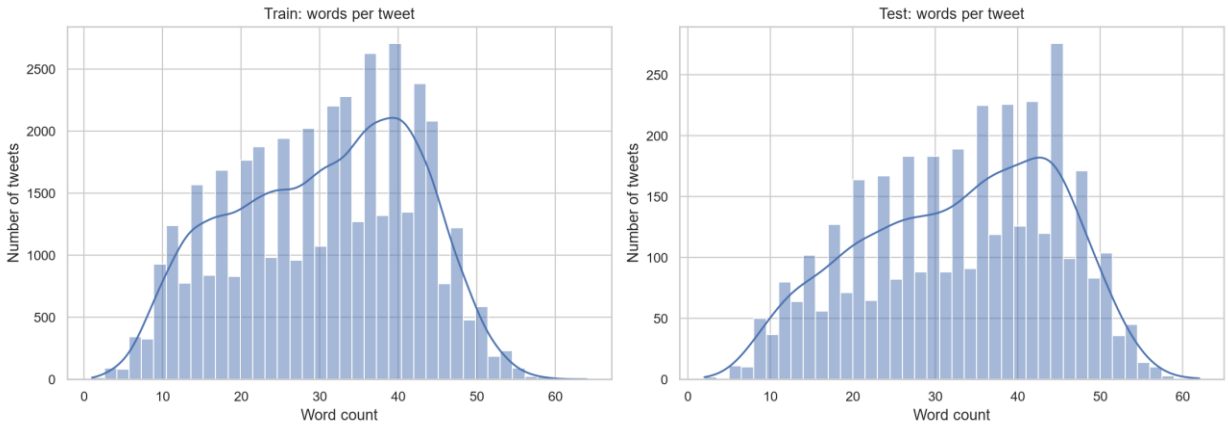
| Statistic | Training set | Test set |
|---|---|---|
| Mean characters per tweet | 204.20 | 213.44 |
| Median characters per tweet | 215 | 227 |
| Mean words per tweet | 30.50 | 32.91 |
| Median words per tweet | 32 | 34 |
| Mean URLs per tweet | 0.58 | 0.48 |
| Mean hashtags per tweet | 1.97 | 2.06 |
| Mean mentions per tweet | 0.43 | 0.32 |




The training tweets contain between 1 and 64 whitespace-separated words, while the test tweets contain between 2 and 62. Tweets also contain URLs, hashtags, user mentions, punctuation, and other informal language patterns commonly found in social media text.

These characteristics make the dataset suitable for evaluating models that learn semantic and contextual information from text. Preprocessing will be kept consistent between training, validation, and testing, while avoiding unnecessary removal of sentiment-bearing information such as negation, punctuation, or emotive expressions.

### 3.3. Text Normalization and Padding Strategy



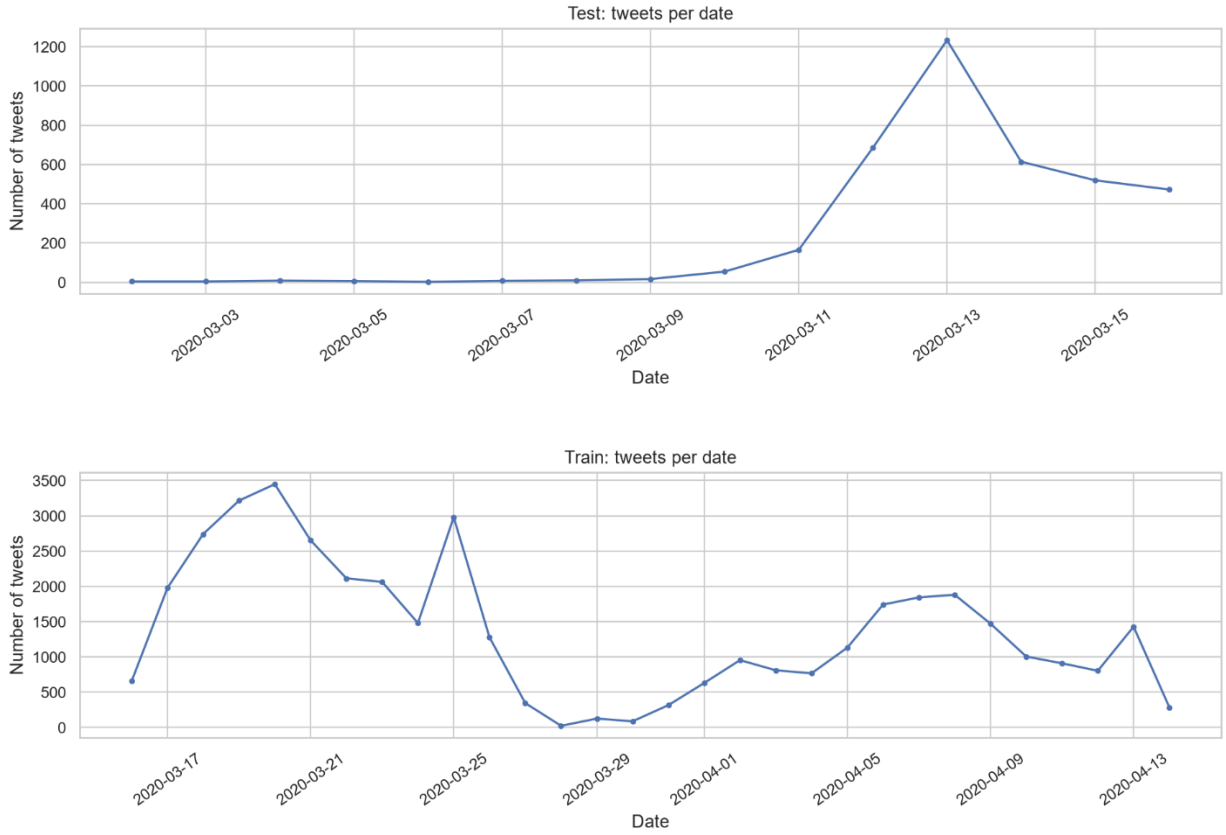

Text preprocessing will be applied consistently to the training, validation, and test sets. Tweets will be checked for missing or empty text, and normalization will be kept minimal to preserve sentiment-related information, including negation, punctuation, and hashtags. URLs, user mentions, and redundant whitespace will be handled consistently before tokenization. Based on the preliminary dataset analysis, tweets contain an average of 32.91 words and 213.44 characters, with a maximum of 62 words and 342 characters in the supplied test set. Each tweet will be converted into a sequence of token IDs and padded to a fixed maximum sequence length determined from the training data. Shorter sequences will be padded with a designated padding token, while sequences exceeding the selected limit will be truncated. Padding tokens will be excluded from meaningful model computation using attention masks or appropriate sequence-length handling, depending on the model architecture.

### 3.4. Duplicate and Leakage Examination

The preliminary examination found 15 repeated normalized tweet texts within the training set and no repeated normalized texts within the test set. One normalized text appears in both splits, corresponding to approximately 0.03% of the unique test texts. No normalized tweet was found with conflicting sentiment labels across the combined dataset. The UserName and ScreenName identifier values also do not overlap between the two files.

The single train–test text overlap will be inspected before final evaluation. If it represents a genuine duplicate, its implications will be documented. The supplied test split will otherwise be preserved to maintain comparability with the dataset's original organization.

### 3.5. Temporal Distribution




The training set covers 16 March to 14 April 2020, while the test set covers 2 March to 16 March 2020. The two files therefore have different temporal distributions, with the test data primarily preceding the training data.

This is an important dataset characteristic because COVID-19 discussions and public sentiment may change over time. Performance on the supplied test set may consequently reflect temporal distribution differences in addition to classification difficulty. The original split will be retained for the primary evaluation, and this limitation will be discussed when interpreting the results.

### 4. Data Preparation and Split Strategy

The supplied training and test files will be used as follows:

- **Training data:** The 41,157 samples in Corona_NLP_train.csv.
- **Validation data:** A stratified validation subset created from the training file, with the remaining samples used for model fitting.
- **Test data:** The 3,798 samples in Corona_NLP_test.csv, reserved for final evaluation.

A stratified 90:10 split of the supplied training data is planned, resulting in approximately 37,041 training samples and 4,116 validation samples. The split will use a fixed random seed for reproducibility and will preserve the approximate sentiment-class proportions.

Before splitting, exact and normalized text duplicates will be examined so that duplicate records do not unintentionally appear across training and validation subsets.

Text preprocessing will include handling empty or malformed text if encountered, tokenization appropriate to each model, and consistent truncation or padding to a selected maximum sequence length. Any normalization of URLs, mentions, or other text elements will be specified and applied consistently. The pretrained tokenizer will be used for the transformer model.

## 5. Planned Models

### 5.1. Baseline Model: BiLSTM Text Classifier

The baseline will be a neural text classifier trained from scratch. Tweets will be tokenized and converted into token indices, which will be mapped to trainable word embeddings. A bidirectional LSTM (BiLSTM) will encode the token sequence, and a classification head will predict one of the five sentiment categories.

The model will provide a meaningful deep-learning baseline without relying on pretrained language-model weights. Its performance will establish a reference point for assessing the benefits of contextual representations learned from large-scale text.

### 5.2. Modern Model: Pretrained DistilBERT

The modern model will be DistilBERT pretrained on general-language corpora and fine-tuned for the five-class sentiment classification task. Its original language-model representation will be adapted to the task by adding or replacing the classification head with a five-class output layer.

Fine-tuning will update the model parameters using the training data, with the validation set used to select hyperparameters and the best checkpoint. The experiment will record the selected pretrained checkpoint, tokenizer, maximum sequence length, learning rate, batch size, number of epochs, and early-stopping or checkpoint-selection criterion.

DistilBERT is selected as a relatively compact transformer model that can capture contextual relationships in text while requiring less computation than larger transformer architectures.

## 6. Controlled Experiment

**Research question:** Does fine-tuning a pretrained contextual language model improve COVID-19 tweet sentiment classification compared with a BiLSTM model trained from scratch?

**Hypothesis:** DistilBERT will achieve a higher macro-F1-score than the BiLSTM baseline because its pretrained contextual representations provide stronger semantic features for distinguishing closely related sentiment categories.

**Independent variable:** Model architecture and initialization strategy: BiLSTM trained from scratch versus pretrained DistilBERT fine-tuned on the task.

**Controlled factors:** Both models will use the same training and validation records, the same target labels, the same untouched test set, and the same evaluation metrics. Validation-based model selection will follow a predefined criterion for each model. Training configurations will be documented to make the comparison reproducible.

**Primary decision criterion:** Compare test macro-F1-score after selecting checkpoints and hyperparameters using only the training and validation data. Accuracy, per-class F1-scores, and the confusion matrix will provide supporting evidence.

The experiment will also examine errors between Negative and Extremely Negative, and between Positive and Extremely Positive. These neighboring categories may be difficult to distinguish because sentiment intensity can be expressed subtly in informal text.

**7. Evaluation Metrics and Error Analysis**

The primary metrics will be:

- **Accuracy:** The proportion of test samples assigned the correct sentiment label.
- **Macro-F1-score:** The unweighted mean of the F1-scores across all five sentiment classes.
- **Per-class precision, recall, and F1-score:** To identify classes for which the model performs poorly.
- **Confusion matrix:** To examine systematic confusion between sentiment categories.

Error analysis will include representative misclassified tweets, with particular attention to negation, sarcasm, mixed sentiment, ambiguous wording, and differences in sentiment intensity. The analysis will also consider whether URLs, hashtags, mentions, or temporally specific expressions affect predictions.

The report will avoid treating high overall accuracy alone as evidence of robust performance. Results will be interpreted alongside per-class metrics and the observed temporal differences between the supplied training and test sets.

**8. Compute-Cost Analysis**

Both models will be trained using the same available computing environment where practical. The following will be recorded:

- Number of trainable parameters and total parameters, where applicable.
- Training time per epoch and total training time.
- Peak GPU memory usage, if a GPU is available.
- Inference latency or throughput on a fixed number of test samples.
- Batch size, maximum sequence length, optimizer, and learning rate.
- Number of epochs and the checkpoint-selection procedure.

The BiLSTM is expected to be computationally lighter than DistilBERT, while DistilBERT may provide stronger classification performance. The actual trade-off will be measured rather than assumed.

**9. Expected Outcome and Limitations**

This project is expected to establish a reproducible comparison between a neural sequence model trained from scratch and a pretrained transformer fine-tuned for COVID-19 tweet sentiment classification. The comparison will quantify whether contextual pretraining improves classification quality and whether the improvement justifies the additional computational cost.

Several limitations will be acknowledged. First, the sentiment labels are provided by the dataset and may contain annotation ambiguity. Second, social-media language includes sarcasm, slang, and context-dependent expressions that remain challenging for automated classifiers. Third, the training and test sets cover different time periods, potentially introducing temporal distribution differences. Finally, conclusions drawn from COVID-19-related tweets may not generalize to unrelated topics or other forms of text.

**10. Summary**

The proposed task uses 41,157 training samples and 3,798 test samples to study five-class sentiment classification on COVID-19-related tweets. Preliminary examination indicates complete tweet text and label fields, moderate class imbalance, informal text with frequent hashtags and URLs, minimal normalized text overlap between the supplied splits, and a temporal difference between training and testing.

The project will compare a BiLSTM baseline trained from scratch with a fine-tuned pretrained DistilBERT model. Accuracy and macro-F1-score will be the principal evaluation metrics, supplemented by per-class analysis, confusion matrices, error analysis, and compute-cost

measurements. This design provides a clear experimental comparison while accounting for the dataset's characteristics and limitations.